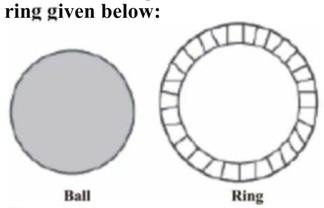
PHYSICS

87. Consider the figures of a metal ball and a metal ring given below:



The metal ball can just pass through the hole of a metal ring formed out of a strip. When the ball is heated it gets stuck. But when the metal ring is heated.

(a) The ball can still pass through it because the ring diameter expands on heating.
(b) The ball gets stuck because the diameter of the hole decreases on expansion.
(c) The ball will still pass through because the hole diameter does not change.
(d) The ball will pass through because there is no change in the ring.

Ans. (a) : The metal ball can just pass through the hole of a metal ring formed out of a strip. When the ball is heated it gets stuck. But when the metal ring is heated the ball can still pass through it because the ring diameter expands on heating. This happens by virtue of a property of metal as it elongate and expand on heating and contract while cooling.

88. The working of the quartz crystal in the watch is based on the:

(a) Photoelectric Effect (b) Johnson Effect
(c) Piezoelectric Effect (d) Edison Effect

Ans. (c) : Quartz crystal clock based on piezo-electric effect, it is an oscillating EMF develops in one pair of faces of the crystal when a variable pressure is applied on another pair of faces of the crystal perpendicular to the first one.

(Source - 12th NCERT)

89. At what temperature are the readings of a Centigrade and Fahrenheit thermometer the same?

(a) -40 (b) 212
(c) 40 (d) 100

Ans. (a) : This is the most common temperature scale in the world and the simplest to understand. Interestingly, centigrade is a name for Celsius. The freezing point of water for Celsius is 0°C, whereas the boiling point of water is 100°C.
The freezing point of water for Fahrenheit is 32°F, whereas the boiling point of water is 212°F. The abbreviate of Fahrenheit is °F.
The Conversion from Celsius to Fahrenheit and vice versa can be done as
°C = °F – 32 x (5/9)
°F = °C / (5/9) + 32
At, -40 temperature the readings of a Centigrade and Fahrenheit thermometer will be the same.

90. The formation of brilliant colours in soap film is a consequence of the phenomenon of:

(a) Multiple reflection and interference
(b) Multiple refraction and dispersion
(c) Diffraction and dispersion
(d) Polarization and interference

Ans. (a) : The colours seen in a soap bubble arise from interference of light reflecting off the front and back surfaces of the thin soap film. The brightest colors are those that interfere constructively. These rays will interfere in a way that depends on the thickness of the film and the indices of refraction of the various media.

CHEMISTRY

91. Which one of the following is used for dating archaeological specimens in wood, bones and shells?

(a) Uranium 238 (b) Argon isotope
(c) Carbon 14 (d) Strontium 90

Ans. (c) : The Carbon isotope also is used as a tracer in following the course of particular carbon atoms through chemical or biological transformations. In carbon-14 dating, measurements of the amount of carbon-14 present in an archaeological specimen, such as a tree, are used to estimate the specimen's age.

92. Acid precipitation is now regarded as a serious problem in some European and Asian countries. Its major cause or source is:

(a) Discharge of acidic effluents onto neutral or slightly alkaline land where the acidic components precipitate.
(b) Emissions of sulphur oxides and nitrogen oxides from thermal power plants and burning of fossil fuels; these oxides dissolve in atmospheric water vapour and fall back on earth as acid rain.
(c) Natural carbon dioxide released during respiration of living organisms dissolves in water, forming carbonic acid which is the chief contributor to acidity in rain water.
(d) Chlorofluorocarbons readily react with various chemicals near the earth's surface, producing acidic intermediates which cause acid precipitation.

Ans. (b) : Acid precipitation is now regarded as a serious problem in some European and Asian countries. Its major cause or source is emissions of sulphur oxides and nitrogen oxides from thermal power plants and burning of fossil fuels; these oxides dissolve in atmospheric water vapour and fall back on earth as acid rain.
Acid rain results when sulfur dioxide (SO_2) and nitrogen oxides (NO) are emitted into the atmosphere and transported by wind and air currents. The SO_2 and NO react with water, oxygen and other chemicals to form sulfuric and nitric acids. These then mix with water and other materials before falling to the ground.

93. Which one of the following colloidal systems is represented by fog?

(a) Liquid in gas (b) Gas in liquid
(c) Solid in gas (d) Liquid in liquid

Ans. (a) : A colloidal system is a type of mixture in which one part is dispersed constantly throughout another. Colloidal systems are usually formed when one part is dispersed through another, but does not combine to form a solution.
Fog is an example of an aerosol particle thus cloud or fog is a colloidal system in which the dispersed phase is liquid and dispersion medium is gas. Therefore the correct answer to the question cloud or fog is a colloidal system in which the dispersed phase and dispersed medium are liquid and gas.

94. The Government of India has totally banned the export of 'Burnt Coal' (Jhama), because it contains recoverable amount of a metal/element used in transistors. Which one is the element?

(a) Phosphorus (b) Germanium
(c) Silicon (d) Tungsten

Ans. (c) : Depending on the coal that was burned, the chemical composition found in coal ash can vary. However, most of coal ash contains aluminium oxide (Al_2O_3), calcium oxide (CaO) and silicon dioxide (SiO_2) which is very harmful for humans.
Silicon is a relatively non reactive element but can react at high temperature (molten state). Silicon acts like a metal and non metal which enables it to be used in electronics and can be used in extremely precise applications like semiconductors.

95. Assertions (A): A mixture of salt and ice gives temperature below 0°C.
Reason (R): The salt raises the freezing point of ice.
In the context of the above two statements, which one of the following is correct?

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (c) : A mixture of salt and ice gives temperature below 0°C and thus assertion A is true, whereas the salt mixed with ice decreases freezing point of ice and thus reason R is false.

96. Vinegar is an aqueous solution of:

(a) oxalic acid (b) citric acid
(c) acetic acid (d) hydrochloric acid

Ans. (c) : Vinegar is an aqueous solution of acetic acid and trace compounds that may include flavorings. Vinegar typically contains 5–8% acetic acid by volume. Usually, the acetic acid is produced by a double fermentation; converting simple sugars to ethanol using yeast and ethanol to acetic acid by acetic acid bacteria.

97. A fertile soil, suitable for growing common crops, is likely to have a pH value of:

(a) Three (b) Four
(c) Six to seven (d) Nine to ten

Ans. (c) : It has been determined that most plant nutrients are optimally available to plants within this 6.5 to 7.5 pH range, plus this range of pH is generally very compatible to plant root growth.

BIOLOGY

98. The theory of 'jumping genes' was propounded by:

(a) Gregor Johann Mendel
(b) Thomas Hunt Morgan
(c) Barbara McClintock
(d) Watson and Crick

Ans. (c) : Transposable elements (TEs), also known as "jumping genes," are DNA sequences that move from one location on the genome to another. These elements were first identified more than 50 years ago by geneticist Barbara McClintock of Cold Spring Harbor Laboratory in New York.

99. Camel uses its hump for:

(a) Storing water
(b) Storing fat
(c) Balancing the body during walking in desert sand
(d) Temperature regulation

Ans. (b) : A camel's hump is used for storing fat. The camel uses it as nourishment when food is scarce. If a camel uses the fat inside the hump, the hump will become limp and droop down.

100. Poison glands of snakes are homologous to:

(a) Electric organs of fishes
(b) Stings of rays
(c) Sebaceous glands of mammals
(d) Salivary glands of vertebrates

Ans. (d) : Poison glands of snakes are homologous to salivary glands of vertebrates.

101. Which one of the following seeds can benefit a patient of diabetes mellitus by normalizing his blood sugar level?

(a) Coriander seeds (b) Mustard seeds
(c) Cumin seeds (d) Fenugreek seeds

Ans. (d) : Diabetes is a disease that occurs when your blood glucose, also called blood sugar, is too high. Blood glucose is your main source of energy and comes from the food you eat. Insulin, a hormone made by the pancreas, helps glucose from food get into your cells to be used for energy.
Fenugreek seeds can benefit a patient of diabetes mellitus by normalizing the blood sugar level.

102. Hemophilia is a hereditary disease carried by:

(a) Females and expressed by females
(b) Females and expressed by males
(c) Males and expressed by females
(d) Males and expressed by males

Ans. (b) : Hemophilia is caused by a mutation or change, in one of the genes, that provides instructions for making the clotting factor proteins needed to form a blood clot. This change or mutation can prevent the clotting protein from working properly or to be missing altogether. These genes are located on the X chromosome. So, it is expressed by males and carried by females.